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DEVELOPMENT ARTICLE

A scoping review of empirical research on AI literacy assessments

Jiahong Su[1] · Xinyu Chen[1] · Samuel Kai Wah Chu[2] · Xiao Hu[1]

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Abstract
AI literacy is becoming increasingly popular in education, yet there has been limited focus on reviewing empirical research on AI literacy assessment. The purpose of this study was to synthesize existing empirical studies on AI literacy assessment, with the aim to understand how AI literacy has been assessed and to inform future AI literacy assessment development. This scoping review evaluates and synthesizes 36 studies on AI literacy assessment published between 2019 and 2024, involving assessment tools, forms of assessment, and reliability and validity evidence related to AI literacy assessment. The review proposes four aspects (i.e., knowledge of AI, AI ethics, affect towards AI, and use of AI) for assessing AI literacy. The results showed that (1) most research focused on assessing primary and secondary school students' AI literacy; (2) most studies used questionnaires, followed by surveys; (3) most studies used computer-based tests, followed by paper-based tests; (4) most studies assessed participants' AI knowledge, followed by AI ethics; and (5) only a few studies reported evidence of the reliability and effectiveness of their assessments. Based on the reviewed literature, this study develops an AI literacy framework for people of all ages and from all countries. The findings and directions for future research are also discussed.

Keywords Artificial intelligence literacy · AI literacy · AI literacy framework · Assessment · Scoping review

Introduction

Artificial intelligence (AI) literacy has gained significant attention in education since it was promoted by Ng et al. (2021) and Casal-Otero et al. (2023). Long and Magerko (2020) defined *AI literacy* as a set of competencies that are "*necessary in a future in which AI transforms the way that we communicate, work, and live with each other and with machines"* (p. 1). Studies regarded AI literacy as the fundamental ability required to become an independent citizen in the AI era (e.g., Yi, 2021).

✉ Jiahong Su
maggiesu@connect.hku.hk

1 Faculty of Education, The University of Hong Kong, Hong Kong SAR, China

2 School of Nursing and Health Studies, Hong Kong Metropolitan University (HKMU), Hong Kong SAR, China

This study aims to systematically review AI literacy assessment research by summarizing the existing evidence regarding assessment tools, forms of assessment, and reliability and validity to assess AI literacy across different application domains, providing a quick reference for researchers interested in conducting AI literacy assessment, and providing evidence of the assessment methods used to measure AI literacy. We believe that understanding the basic concepts, principles, and applications of AI is crucial for the general public and students as AI has permeated every aspect of our lives. This helps them better comprehend the impact of AI technology on society and personal life, as well as how to make better use of AI technology. Therefore, this research is important for the general public and students. This study focuses on students and the general public rather than AI experts because educating the general public about AI more broadly can increase societal acceptance and integration of AI technologies. Additionally, a wider understanding of AI can help mitigate potential risks and ensure that AI technologies are developed and used responsibly. This study also develops an AI literacy framework that is suitable for people of all ages and from all countries.

Literature review

Definition and significance of AI literacy

Ever since MacCarthy first described AI as "the science and engineering of making intelligent machines" (MacCarthy, 2007, p. 2), the concept of AI has raised considerable attention across the fields of science, technology, business, and education. The basic objective of AI is to understand, simulate, and imitate human cognitive functions and intelligence into an artificial system (Casal-Otero et al., 2023). The origins of the term "AI literacy" may be traced back to an online publication that was released in 2015 (Konishi, 2015), subsequently referenced by an empirical research study (Kandlhofer et al., 2016). Some defined AI literacy as the skills and abilities to understand, evaluate and make effective use of artificial intelligence (AI) in a variety of settings, including but not limited to online communities, homes, and workplaces (Long & Magerko, 2020). Like many other literacies in the technology field (e.g. Data Literacy, Visual Literacy, Media Literacy), AI literacy is proposed to indicate the understanding and proficiency in certain technological concepts for not only computer experts, but also the general public (Laupichler et al., 2022a).

More and more attention in recent years has been raised on the need of enhancing the rate at which people from various professional backgrounds can acquire the requisite skills and knowledge for positions in a workforce that incorporates AI (Manyika & Sneader, 2018). The significance lies in the fact that individuals who experience job displacement due to the advent of AI and other forms of automation are required to cultivate their AI literacy in order to facilitate their reintegration into the workforce (Faruqe et al., 2021). In the House of Lords Artificial Intelligence Select Committee's report (UK), it is clearly stated that the influence of AI on future generations is inevitable, regardless of how quickly or slowly it will advance (House of Lords, 2018). For young people to thrive in a world dominated by AI, they will need to acquire a wide variety of skills linked to AI and its development (Wong et al., 2020). A growing number of occupations need knowledge of AI, leading many to argue that AI literacy is just as crucial as reading and writing literacy (Kandlhofer et al., 2016).

AI literacy frameworks

Several frameworks have been proposed to understand and prompt AI literacy for students and also the general public. In 2019, The "AI for K12" team has established five AI "big ideas" to direct the advancement of educational standards (Touretzky et al., 2019), which built a robust groundwork for further studies on the promotion of AI literacy. Later, after reviewing a wide range of literature from different fields, Long and Magerko (2020) laid up a thorough framework for AI literacy, including 17 core competencies. Further, Wong et al. (2020) have suggested that AI literacy in K-12 education should center on three areas: AI applications, AI concepts, and AI ethics/safety. One AI literacy framework developed by Kong et al. (2021) includes three dimensions: Cognitive, Affective and Sociocultural dimensions. Teaching students how to analyze and comprehend the actual world via the lens of AI is the focus of the Cognitive component. The objective of the Affective component is to provide individuals with the confidence to engage fully in the digital world. And the ethical use of AI is a key issue in the sociocultural domain. Although ethical concerns with AI have gained more attention, it has not been a prominent part of AI courses, and there is little information on the ethical topics addressed in AI classes (Kong et al., 2023a, 2023b).

Ng et al. (2021) reviewed thirty articles and concluded with an AI literacy framework which has four key components used to foster AI literacy: "Know and understand AI", "Use and apply AI", "Evaluate and create AI", and "AI ethics". First, "know and understand AI" refers to understanding fundamental concepts, proficiencies, and knowledge pertaining to AI, which aims to enable individuals from diverse backgrounds to get a thorough comprehension of AI. Second, "use and apply AI" means to provide learners with the information and abilities needed to successfully apply AI concepts in many settings and everyday life scenarios. Third, "evaluate and create AI" refers to the capacity to critically evaluate AI technology, alongside the aptitude to use AI for efficient communication and collaboration, with critical thinking skill and problem-solving ability. Creating AI means the ability to design, assemble, comstruct, build and develop new AI applications. AI ethics means to educate both citizens and computer scientists with human-centered values related to AI. According to the ethical principles elaborated on in Kong et al. (2021), AI should: (1) respect human autonomy; (2) have benefits that outweigh drawbacks; and (3) balance advantages and risks. Recent ethical frameworks for AI have also included the three concepts of autonomy, beneficence and justice as outlined by Floridi and Cowls (2022). It is crucial for enhancing corporate social responsibility, taking diversity and inclusion into account while using AI for the benefit of society (Ng et al., 2021).

Nevertheless, Faruqe et al. (2021) pointed out the existing AI literacy frameworks remain at a preliminary stage, lacking comprehensive development to effectively cater to diverse contexts such as educational environments, job fairs, and promotion protocols.

AI literacy assessment

The development and study of AI literacy assessment is critical at this moment due to the growing reliance on AI skills across various domains. As AI becomes pervasive in education, healthcare, and governance, there is a need to ensure that individuals possess not only technical knowledge but also critical thinking and ethical reasoning skills to navigate AI-driven systems responsibly (Ng et al., 2021). Assessment not only drives learning and instruction—"what is assessed is taught" (Johnston & Costello, 2005)—but also provides

insights into the effectiveness of AI literacy programs. Without robust assessment tools, it is challenging to measure the impact of educational interventions or identify gaps in learners' understanding (Laupichler et al., 2022a). Moreover, assessments play a crucial role in promoting equity, allowing educators and policymakers to design targeted interventions for underrepresented groups (OECD, 2012).

Designing an AI literacy assessment tool might help clarify the concept of AI literacy by requiring a structured approach to the issue (Laupichler et al., 2022a). For example, the development of the Artificial Intelligence Literacy Scale (AILS) aimed to assess non-expert adult users' self-reported competencies in utilizing AI technologies (Wang et al., 2023). This scale offers a framework for assessing people's perceived AI engagement skills, providing information on how prepared they are to engage with AI-driven systems. Similarly, the Meta AI Literacy Scale (MAILS) was designed to measure various facets of AI literacy for adults, including understanding, application, detection, and ethical considerations (Carolus et al., 2023). By encompassing a broad spectrum of AI-related competencies, MAILS facilitates a comprehensive assessment of an individual's proficiency in navigating AI environments. Furthermore, the AI Literacy Questionnaire (AILQ) was developed to evaluate affective and cognitive dimensions of AI literacy among secondary students (Ng et al., 2024). This tool facilitates the comprehension of students' perceptions of AI and their knowledge of AI concepts, therefore guiding the development of educational interventions that promote successful AI learning experiences. Overall, aligning AI literacy assessment with a growth mindset is crucial for fostering resilience and a positive attitude toward learning, especially given the rapid evolution of AI technologies. A growth mindset helps learners to view challenges as chances for development, and diminishes anxiety and fear of failure (Casal-Otero et al., 2023). Incorporating a growth mindset into AI literacy exams involves developing evaluation methods that evaluate not only existing skills but also fostering continuous learning and improvement. For example, continuous feedback via formative assessment supplants high-stakes examinations, which can develop well-rounded and flexible skills that will allow individuals to complement AI systems and keep pushing their innovations (George, 2023).

Research gaps

First, the field of AI literacy study is still in its early stages, especially with little availability of literature on its assessments. Casal-Otero et al. (2023) pointed out in their systematic review that there is a scarcity of AI-related educational studies that have conducted assessments on learning objectives, particularly students' understanding of AI concepts. Second, current reviews on AI literacy lacked a comprehensive analysis on assessment perspective. Ng et al. (2021) listed knowledge tests, survey, portfolio assessment and interviews as three main types of assessment tools used in AI literacy literature, but they did not look further into the specifics of these evaluations in their application settings or trustworthiness and accuracy. Similarly, Laupichler et al. (2022b) also listed evaluation and assessment methods used in past literature without any further analysis. Here, we propose a scoping review specifically on AI literacy assessments from empirical research which will not only provide researchers and educators with detailed analysis on various choices of current existing assessments methods and tools, but also shed light on the design of future AI literacy assessments and studies, hence to prompt the development of AI literacy theories.

Research objectives and questions

In this study, we conducted a scoping review to evaluate and synthesize relevant literature on AI literacy assessments. This study only reviews studies investigating AI literacy of students and the general public, rather than that of AI researchers or AI developers. The objective is to examine the existing assessment tools, forms of assessment, and reliability and validity of evidence ascribed to the existing research literature on AI literacy assessments.

The scoping review was guided by the four research questions (RQs):

RQ1: What are the characteristics of the empirical studies on AI literacy assessments?
RQ2: What constructs are measured by the AI literacy assessments?
RQ3: What assessment tools are used to assess AI literacy?
RQ4: What is the reliability and validity evidence of these AI literacy assessments?

Methods

Literature search

The literature search was initiated by querying two electronic databases: Web of Science and Scopus. These two databases were selected based on their authoritative status and frequent use in conducting literature reviews in education research (Su et al., 2023a, 2023b).

We initially gathered all journal articles across various fields that exclusively featured the term “AI literacy” in their titles. Despite the association of AI literacy with AI in education (AIED), we refrained from searching for these terms as alternative keywords. This decision was made based on our belief that AI literacy is a distinct concept from AIED, and our focus was solely on researchers acknowledging AI literacy and using the term "Artificial Intelligence literacy.”

Regarding the reference types, we decided to exclude conference papers from this review for the following reasons: (a) most significant scientific achievements are reported in journal articles. In such cases, conference papers might report similar results to journal articles. (b) most conference papers on AI literacy research lack specific information regarding their assessment tools, which could impact the outcomes of this literature review study. Many of them only designed a course without collecting any data.

We used the following inclusion criteria to select articles: (a) the use of the term “AI literacy” or “Artificial intelligence literacy” in the title of the paper; (b) peer-reviewed journal articles; (c) empirical studies containing assessments of AI literacy; (d) written in English. In the end, as of August 2024, there were 51 articles collected. Subsequently, the first author downloaded all selected articles.

After collecting the articles, the first and second authors jointly reviewed the titles, abstracts, and methods sections, excluding irrelevant articles such as those that did not focus on AI literacy assessment. Discrepancies between the first and second authors were resolved through discussion and further review of contentious studies. The final evaluation reduced the number of articles to 36 (Fig. 1).

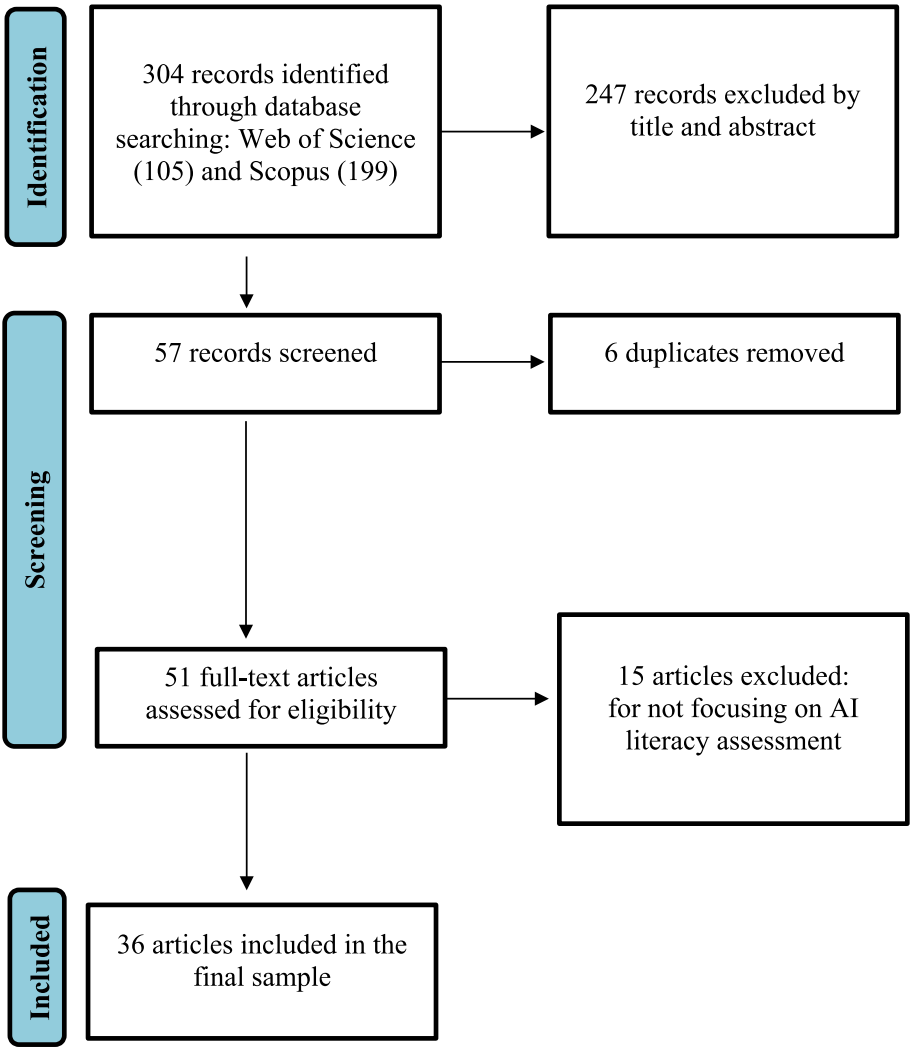

Fig. 1 The search protocol adopted in this scoping review

Analysis

We employed text classification to code the literature following the content analysis procedure. Initially, we developed a coding scheme to systematically extract information from each selected article based on the four research questions. An Excel spreadsheet was used to store and analyze all data. Subsequently, using the initial coding scheme, the first and second authors independently reviewed and coded eight randomly selected articles. Inter-rater reliability across all categories reached 90%. Any discrepancies were resolved through discussion with the second author. Finally, given the acceptable

inter-rater agreement, the first author independently coded the remaining articles. The primary coding results are presented in Appendix A.

Results

RQ1: What are the characteristics of the empirical studies on AI literacy assessments?

The number of publications on AI literacy assessments continues to grow, increasing from one in 2019 to fifteen in 2023. The observed trend in publication growth aligns with Ng's (2021) findings. This evidence supports the notion that research on AI literacy assessments is becoming increasingly important.

Researchers conducted AI literacy assessment studies across various countries and regions. Most studies were conducted in Hong Kong (e.g., Chiu, 2021; Ng et al., 2023; Su & Yang, 2023a, 2023b), followed by mainland China and the United States (e.g., Cardon et al., 2023; Zhao et al., 2022). Other participants' countries or regions include Finland, India, Israel, Korea, Germany, Spain, Taiwan, and Belgium (e.g., Carolus et al., 2024; Yuan et al., 2024). However, two articles mention multiple countries of the participants (e.g., Laupichler et al., 2023a, 2023b).

Most studies (75%) developed an AI literacy assessment to assess K-16 students' AI literacy. Among these, 77% focused on evaluating primary and secondary school students. The rest integrated AI literacy into early childhood, teacher education programs, and adults (25%). In general, a significant amount of research has been dedicated to assessing AI literacy in K-16 schools, far more than in teacher education or other adult practices.

RQ2: What constructs are measured by the AI literacy assessment?

Knowing & understanding AI (n = 31), AI ethics (n = 12), affect towards AI (n = 8), and using AI (n = 4) in AI literacy assessment are often measured in the selected studies (Table 1). It is noteworthy that one study may address multiple constructs.

Table 1 Constructs being measured in AI literacy assessment

Construct	N	Definitions	Sample studies
Know & understand AI	31	Know the basic knowledge of AI	Vartiainen et al. (2020); Chiu (2021)
AI ethics	12	Human-centered considerations, such as fairness, accountability, ethics, and safety	Dai et al. (2023a); Kong et al. (2023b)
Affect towards AI	8	Attitudes/motivation toward AI	Chiu (2021); Su and Yang (2023b)
Use AI	4	Be able to use AI applications	Cardon et al. (2023); Zhao et al. (2022)

Knowing and understanding AI

Thirty-one articles focused on instruments assessing participants' AI knowledge (e.g., Chiu, 2021; Kong et al., 2023a) (see Table 2). The tools used to assess this construct included surveys (12 studies), tests (11 studies), questionnaires (9 studies) and interviews (6 studies). It is noteworthy that one study may employ multiple assessment tools to evaluate participants' understanding of AI. Although there were some overlaps between the assessment methods, this paper only directly quotes the terms used in the review studies.

Most studies used surveys to assess participants' understanding of AI knowledge (e.g., Mertala et al., 2022; Priya et al., 2022). Priya et al. (2022) developed a survey to assess K-12 students' understanding of machine learning concepts, namely supervised learning, gradient descent, and K-Nearest Neighbor (KNN). Ten studies used tests to assess participants' AI knowledge (e.g., Kong et al., 2023a; Su, 2024). Kong et al. (2023a) developed an AI literacy concepts test to assess university students' AI literacy. Ding et al. (2024) developed a test using true/false and multiple-choice questions to measure participants' AI knowledge. Nine studies used questionnaire to assess participants' understanding of AI (e.g., Carolus et al., 2023). Carolus et al. (2023) developed a self-assessment questionnaire to assess participants' understanding of AI, such as "*I know the definitions of AI*". Overall, surveys, tests and questionnaire can be used to assess participants' AI knowledge. Some researchers used self-assessment in questionnaires to measure participants' understanding of AI (Laupichler et al., 2024), such as "*I can tell if the technologies I use are supported by artificial intelligence*".

AI ethics

Twelve articles focused on using assessment tools to assess participants' understanding and awareness of AI ethics (Dai et al., 2023a; Kong et al., 2023b) (see Table 2). The assessment tools used to assess the AI ethics construct in the AI literacy assessment research include questionnaire (5 studies), survey (3 studies), self-reflective writing (2 studies), and test (1 study). Carolus et al. (2023) developed a self-assessment questionnaire to measure participants' understanding of AI ethics, such as "*I can weigh the consequences of using AI for society*". Kong et al. (2023b) used an AI ethics survey and self-reflective writing on AI ethics to assess secondary school students' AI ethics. The AI ethics survey includes 12 items to measure the secondary school students' understanding of AI ethical issues, including human autonomy, the beneficence of AI, and fairness of AI (Kong et al., 2023b). Secondary school students must write down their views on AI ethical considerations in English and Chinese in self-reflective writing on AI ethics assessment (Kong et al., 2023b). Dai et al. (2023a) developed multiple-choice question (MCQ) items to assess upper primary school student's ethical awareness of AI. Ethical awareness was assessed using multiple-choice questions (MCQ) in a paper-and-pencil test.

Table 2 Constructs and tools of AI literacy assessment

Studies	Constructs	Assessment tools
Vartiainen et al. (2020)	Know & understand AI	Video recordings and interviews
Chiu (2021)	Know & understand AI; Affect towards AI; AI ethics	Questionnaire
Priya et al. (2022)	Know & understand AI	Survey
Shamir and Levin (2022)	Know & understand AI; Affect towards AI	Questionnaire
Dai et al. (2023a)	Know & understand AI	Pre- and post-interviews
Wood et al. (2021)	Affect towards AI	Questionnaire
Laupichler et al. (2023a)	Know & understand AI	Questionnaire
Choi and Park (2021)	Know & understand AI	Survey
Zhao et al. (2022)	Know & understand AI; Use AI; AI ethics	Survey
Laupichler et al. (2023b)	Know & understand AI	Survey
Laupichler et al., (2022a)	Know & understand AI	Questionnaire
Dai et al. (2023b)	Know & understand AI; AI ethics	Drawing task, assessment task, and Scratch project
Laupichler et al. (2023c)	Know & understand AI; Affect towards AI	Questionnaire
Kong et al. (2023a)	Know & understand AI; AI ethics	Tests, surveys, reflective writings and interviews
Kong et al. (2023b)	Know & understand AI; AI ethics	Pre- and post-course tests, surveys and self-reflective writing
Kong et al. (2022)	Know & understand AI	Conceptual tests, surveys and self-reflective writing tasks
Kong et al. (2021)	Know & understand AI	AI concepts test and AI literacy survey
Mertala et al. (2022)	Know & understand AI	Survey
Estevez et al. (2019)	Know & understand AI	Questionnaire
Zhang et al. (2022)	Know & understand AI; Affect towards AI; AI ethics	Survey, presentation and interviews
Wang et al. (2023)	Use AI; AI ethics	Questionnaire
Lin et al. (2023)	Know & understand AI	Survey
Henry and Collard (2021)	Know & understand AI	Test, observation and recording
Cardon et al. (2023)	Use AI	Survey
Ng et al. (2022)	Know & understand AI	Interviews and writing artefacts
Ng et al. (2023)	Know & understand AI; Use AI; AI ethics	Test, questionnaire, presentation and interviews
Hornberger et al. (2023)	Know & understand AI	Test
Su and Yang (2023a)	Know & understand AI	Test
Su and Yang (2023b)	Affect towards AI	Questionnaire
Su (2024)	Know & understand AI	Test
Carolus et al. (2023)	Know & understand AI; AI ethics	Questionnaire
Ng et al. (2023)	AI ethics; Affect towards AI	Questionnaire

Table 2 (continued)

Studies	Constructs	Assessment tools
Ding et al. (2024)	Know & understand AI; AI ethics	Test
Laupichler et al. (2024)	Know & understand AI; Affect towards AI	Questionnaire
Yuan et al. (2024)	Know & understand AI; AI ethics	Survey
Zhang et al. (2024)	Know & understand AI	Test

Affect towards AI

Eight articles mainly focused on using assessment tools to assess participants' affect towards AI, such as attitudes and motivations towards AI. The assessment tools used to assess this construct in the reviewed studies, include questionnaires (7 studies) and survey (1 study).

Most studies designed questionnaires to assess participants' attitudes and motivation toward AI, followed by survey (e.g., Chiu, 2021; Laupichler et al., 2023c). Chiu (2021) developed a questionnaire to assess secondary school students' intrinsic motivation to learn AI (AIM). Laupichler et al. (2023c) used questionnaire to assess participants' attitudes towards AI. This test has five items adapted from the Sinderman et al. (2021) study. One study developed an attitudes toward AI survey to assess middle school students' interest in AI (Zhang et al., 2022).

Use AI

Four articles focused on using AI tools to develop AI literacy (e.g., Cardon et al., 2023; Zhao et al., 2022) (see Table 2). Two studies used surveys to assess participants' AI literacy and proficiency with the AI tools used in the studies. For example, Cardon et al. (2023) used an AI-assisted writing tool to develop participant's AI literacy. This study used a survey to ask participants about concepts and the perceived benefits of AI-assisted writing. Zhao et al. (2022) used a self-assessment survey to assess teacher proficiency with AI tools. Other two studies employed questionnaires to measure participants' proficiency with AI tools (e.g., Ng et al., 2024; Wang et al., 2023). Ng et al. (2024) developed a self-assessment questionnaire to assess secondary school students' proficiency with AI tools (e.g., "*I can use AI applications (e.g., sensors, machine learning model trainer) to solve authentic problems.*"). Wang et al. (2023) designed a questionnaire to measure participants' proficiency in utilizing AI technologies to complete tasks.

We also noticed that the authors used different tools to assess participants' AI knowledge, such as surveys, tests, and questionnaires. In this review, most articles (31 studies) focused on assessing participants' AI knowledge. Therefore, most assessment tools were for assessing K-12 students' AI knowledge. Moreover, we also found that limited studies used assessment tools to assess participants' motivation to learn AI (Chiu, 2021). Furthermore, according to review studies, we found that only four studies used assessment tools (e.g., questionnaires and surveys) to measure participants' proficiency with AI

tools. Finally, results showed that researchers used evaluation tools to assess participants' AI ethics knowledge across different age groups, except for the kindergarten stage.

RQ3: What assessment tools are used to assess AI literacy?

It is worth noting that an article may assess multiple aspects; however, we only summarize assessment tools for AI literacy assessment. This summary shows some overlaps between assessment tools, and this paper directly quotes the terms used in the reviewed papers.

While there are variations in the design of AI literacy assessments, the literature encompasses two major types of assessment tools: single-use which refers to using only one assessment tool; and combined-use where studies employed more than one assessment tool to gather multidimensional evidence of AI literacy. The frequency of each assessment tool is detailed in Table 3. Most studies used a questionnaire (12 articles), followed by a survey (8 articles) in AI literacy assessment studies. Five studies used tests in AI literacy assessment research. We found that most assessment tools reviewed were computer-based or paper-based methods (Fig. 2).

In terms of the questionnaire, eleven articles used a questionnaire to assess student's AI literacy (e.g., Chiu, 2021; Shamir & Levin, 2022; Laupichler et al., 2022a). For example, Chiu (2021) designed a questionnaire to assess Hong Kong secondary school students' AI literacy.

In terms of surveys, among the eight articles, six focused solely on designing surveys to assess AI literacy without implementing any interventions (e.g., Hornberger et al., 2023; Wang et al., 2023; Zhao et al., 2022). For example, Zhao et al. (2022) developed a survey to assess Chinese primary and secondary teachers' AI literacy. The survey includes knowing and understanding AI (KUAI), applying AI (AAI), evaluating AI Applications (EAIA), and AI ethics (AIE). These items of this survey were adopted from Long and Magerko's (2020) competence. The items included AI systems, weak and strong AI, risks of AI, recognizing AI, human role in AI, etc. Other articles employed surveys to assess participants' machine learning knowledge andconceptions of AI with interventions that aimed to improve learners' AI literacy skill (e.g., Priya et al., 2022). For example, Priya et al. (2022) designed a survey to evaluate higher secondary school students' machine learning concepts after offering a workshop on machine learning.

Five articles used tests to evaluate students' AI literacy (e.g., Ng et al., 2023; Su & Yang, 2023a; Su, 2024; Zhang et al., 2025). For example, Su (2024) designed a test to assess Hong Kong young children's basic AI literacy. Zhang et al. (2025) designed an AI knowledge test to measure secondary school student's AI knowledge.

Other assessment tools include interview in single-use of assessment tools. Combined assessment tool use includes the following: 1. video recording and interview; 2. test, observation, and recording; 3. test, survey, and reflective writing; 4. drawing task + assessment task + Scratch project; 5. tests and surveys; 6. pre- and post-tests, presentation, and interviews; 7. test, questionnaire, presentation, and interviews; 8. interviews and writing artefacts. For example, Kong et al. (2023a) used AI literacy tests, AI literacy and AI empowerment surveys, and self-reflection on writing to assess university students' AI literacy.

Table 3 Assessment tools

Type	Assessment Tool	Studies	Frequency
Single-use	Questionnaire	Chiu (2021), Shamir and Levin (2022), Laupichler et al. (2023a), Wood et al. (2021), Laupichler et al. (2022a), Laupichler et al. (2023c), Estevez et al. (2019), Wang et al. (2023), Su and Yang (2023b), Ng et al. (2023), Laupichler et al. (2024), Carolus et al. (2023)	12
	Survey	Priya et al. (2022), Choi and Park (2021), Zhao et al. (2022), Laupichler et al. (2023b), Mertala et al. (2022), Lin et al. (2023), Cardon et al. (2023), Yuan et al. (2024)	8
	Test	Hornberger et al. (2023), Su and Yang (2023a), Su (2024), Ding et al. (2024), Zhang et al. (2024)	5
	Interview	Dai et al. (2023b)	1
Combined use	Video recording + interview	Vartiainen et al. (2020)	1
	Test + observation + recording	Henry and Collard (2021)	1
	Test + survey + reflective writings	Kong et al. (2022, 2023a, b)	3
	Drawing task + assessment task + Scratch project	Dai et al. (2023a)	1
	Test + survey	Kong et al. (2021)	1
	Pre- and post-tests + presentation + interviews	Zhang et al. (2022)	1
	Test + questionnaire + presentation + interviews	Ng et al. (2023)	1
	Interviews + writing artefacts	Ng et al. (2022)	1

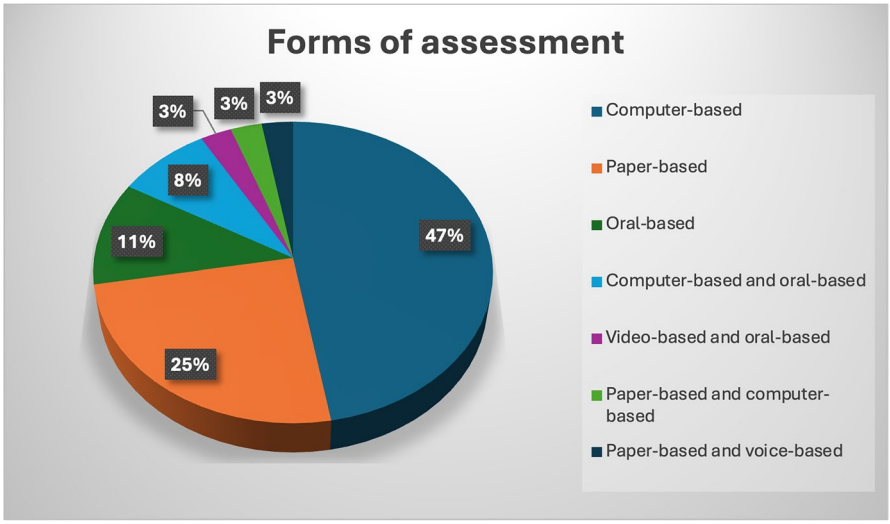

Fig. 2 Forms of assessment

RQ4: What is the reliability and validity evidence of these AI literacy assessments?

Among the 36 studies, 63.8% (23 studies) reported reliability evidence, and less than 20% provided validity evidence. We found that reliability evidence was provided for manual coding, such as inter-rater reliability or Cohen's kappa, especially for knowledge assessment (e.g., Chiu, 2021; Kong et al., 2023b; Su, 2024). Some researchers (Chiu, 2021) used multiple-choice and survey questions to report the internal consistency of tests to determine whether participants obtained similar scores on items measuring the same AI constructs.

Among the studies that reported validity, diverse forms of validity evidence were presented. For example, Zhao et al. (2022) developed an AI literacy test to assess primary and secondary school teachers' AI literacy. This study used two significant methods for the data analysis. The first approach was descriptive and correlation analysis using SPSS software. The second approach was structural equation modeling (SEM) to validate the survey using AMOS software. Laupichler et al. (2023a) developed an AI literacy assessment to assess non-experts' AI literacy. This study used exploratory factor analysis (EFA) for the data analysis.

Although some reliability and validity information has been provided for AI literacy assessments, most AI literacy assessments need more evidence of reliability and validity. Since most studies focus on developing courses for students to learn about AI, the absence of such evidence makes it challenging for the field to confidently employ these assessments in the classroom to measure students' AI learning outcomes.

Discussion

AI literacy is a fast-moving field that researchers have investigated over the past few years (e.g., Sperling et al., 2024; Su et al., 2023a, 2023b). Through this review, we have outlined the current studies of AI literacy assessment implemented across learners of all different ages. In this section, we discuss the main findings and limitations of reviewed studies, based on which a framework of AI literacy assessment is proposed. At the end, we provide future research directions for researchers in the field to enhance their work in AI literacy assessment.

Findings and limitations of reviewed studies

First, most research focuses on primary, secondary, and university students' AI literacy assessment. It is noteworthy that, even though assessing AI literacy development in kindergarten children is challenging due to their limited reading and comprehension abilities (Su et al., 2023a, 2023b; Williams et al., 2024), researchers are trying to implement AI assessments in the early stages of students' cognitive development (Su, 2024). As no conclusive evidence suggests that elementary, secondary, and higher education students are the sole critical stages for students to learn AI, it becomes essential to validate AI literacy assessments for kindergarten children further. By doing so, researchers and practitioners can identify resources that trace students' AI skills development from kindergarten to university.

Second, there was limited research on AI literacy assessment for teacher education and professional development workshops (Su, 2024b). In the review study conducted by Su et al., (2023a), it was found that kindergarten teachers exhibit a lower level of knowledge and confidence in teaching AI compared to traditional kindergarten subjects. Especially considering the limited teacher professional development, introducing AI literacy into the classroom may be a challenge. Ng et al. (2021) also mentioned that introducing AI literacy in the classroom without adequate teacher professional development could be problematic. From this perspective, it is evident that there is an urgent need for more in-depth and systematic research and practices to assist teachers in more effectively implementing AI literacy in classrooms. In this context, it is recommended that targeted professional development plans be developed for teachers to enhance their knowledge and confidence in AI literacy. This plan could be implemented through organizing workshops, providing training courses, and promoting online learning. Additionally, we think there should be an emphasis on strengthening the assessment of teachers' proficiency in AI. For this purpose, various assessment tools can be employed, including teaching observations, knowledge tests, and self-assessments by teachers, to measure their mastery of AI literacy comprehensively.

Third, the most common assessment of AI focuses on evaluating participants' knowledge of AI (i.e., Ng et al., 2021; Su, 2024), with limited consideration for using AI applications. As AI continues to expand across various domains, the evaluation of AI skills may become more comprehensive, encompassing theoretical knowledge and practical applications. Future researchers may consider assessing participants through their involvement in real-world AI projects, such as creating Teachable Machine models, to gauge their ability to apply knowledge and skills in real-world scenarios. Such practical projects can assist participants in translating their knowledge into tangible applications while evaluating their skills in designing, implementing, and assessing AI models. Additionally, these projects can foster participants' creative thinking and problem-solving abilities.

Fourth, we found that most studies (31 studies) used assessment tools to assess participants' understanding of AI, while limited studies (12 studies) focused on using instruments to assess participants' AI ethics and ethical use of AI. CEB (2022) developed principles for the ethical use of AI in the United Nations System, including defining purpose, necessity and proportionality, safety and security, etc. The United Nations (UN) has released a report on a recommendation on the Ethics of Artificial Intelligence in 2022 (United Nations, 2022). This recommendation aims to *provide a basis to make AI systems work for the good of humanity, individuals, societies and the environment and ecosystems, while also prevent harm. It also aims to promote the peaceful use of AI systems* (United Nations, 2022, p. 14). Furthermore, results showed that the researchers evaluated the participants' AI ethics across different ages through surveys, except for the kindergarten stage. At present, AI ethics may not be suitable for kindergarten education. In the future, researchers can use drawing tasks to evaluate young children's views on AI ethics, and the use of storytelling, games and other methods to popularize AI ethics knowledge among young children in a fun way. Simple AI ethics scenarios can be provided for young children to express their views on these situations through drawing, which can help them better understand the importance of AI ethics. This can help young children understand AI ethics, helping them prepare for the future AI era. It is also essential to use appropriate methods and approaches for teaching young children according to their age characteristics and cognitive abilities. For older students or adults, interviews, especially scenario-based ones can be conducted to assess their attitudes toward AI ethics.

From the reviewed studies, we can see that assessing participants' understanding of AI can be measured through methods such as surveys, questionnaires, tests, or interviews (Su, 2024; Vartiainen et al., 2020). However, evaluating aspects such as AI usage can be more difficult due to the involvement of more complex social and human values. Therefore, more comprehensive and systematic approaches such as in-depth interviews and observations are needed to assess these aspects.

Fifth, we also discovered that few studies used assessment tools to assess participants' motivation to learn AI (Chiu, 2021). The aim of evaluating participants' motivation to learn AI is to understand their motives and goals for learning AI and explore their interests and needs in learning AI. This can help educators better design and deliver AI education programs to meet learners' needs and enhance learning outcomes. Additionally, understanding participants' learning motivation can provide valuable information and insights for future research on designing a suitable AI curriculum for learners.

Sixth, we also found that the assessment dimensions for K-12 mainly focused on AI knowledge and attitudes towards AI, while the assessment dimensions for adults were more diverse. In addition to focusing on assessing participants' AI knowledge and attitudes towards AI, the assessment dimensions of adults also included awareness, use, evaluation, and ethics. Moreover, adult assessment dimensions also focused on AI application, authenticity, accountability, and agency. These dimensions aim to comprehensively evaluate adults' cognitive, understanding, and application of AI, as well as their awareness of the ethical and social implications of AI. This means that for adults, the standards for assessing AI literacy are more comprehensive. Adults need to gain a deeper understanding and awareness of the applications and impacts of AI in order to better adapt to the challenges and opportunities of the AI era.

Seventh, most K-12 students are required to participate in a specific course before conducting an AI literacy assessment. These courses use teaching methods such as learning by design, gamification, and flipped classrooms (Choi & Park, 2021; Laupichler et al., 2022a; Su & Yang, 2023b). However, adults, teachers, and university students do not need to take

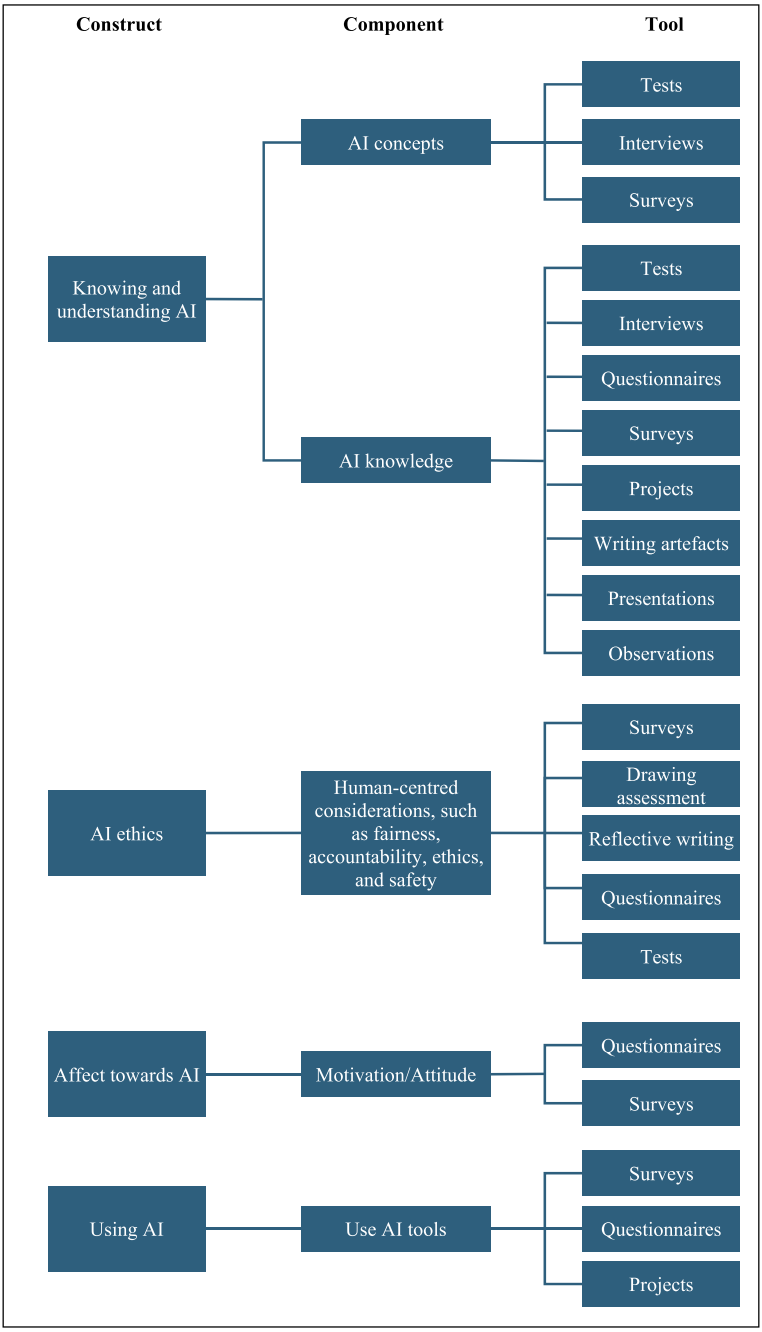

Fig. 3 AI literacy assessment framework

the course before participating in the AI literacy assessment to assess their level of AI knowledge.

Last but not least, reliability and validity evidence of assessment tools should be reported when applicable. As indicators of assessment quality, reliability and validity are crucial for the consistency and accuracy of assessment tools. The results can guide researchers in making necessary revisions to improve existing AI assessments and develop new tools. Additionally, reporting acceptable reliability and validity indices contributes to a broader adoption of the assessment tool. Researchers and educators are more inclined to use assessment tools that have been proven to be reliable and valid.

AI literacy framework

We develop an AI literacy framework based on the results of this scoping review. The framework includes four dimensions: (1) know & understand AI; (2) AI ethics; (3) affect towards AI; (4) use AI, as shown in Fig. 3. We have introduced these four dimensions in the result of RQ2. Future researchers can design AI literacy assessments based on this framework. This framework also can help educators design suitable AI courses for learners. This AI literacy framework is generally targeting people of all ages and from all countries, except that AI ethics might not be suitable for the kindergarten stage at the moment.

Future research directions

AI is a field full of potential. Our review aims to outline the explored landscape of AI literacy assessment. The results indicate that significant work has been done, but more research work is still required. We recommend that researchers and AI assessment developers consider the following future research directions when designing AI assessments.

To begin with, AI can be integrated into all levels of education (e.g., Ng et al., 2022; Su et al., 2023a, 2023b). This research found that AI literacy assessments were developed unequally in the educational environment. They were primarily concentrated in elementary, secondary, and higher education, while their application in the kindergarten domain is relatively limited. Therefore, it is necessary to strengthen research to investigate how AI literacy assessment can be effectively applied in kindergarten settings (Su, 2024). Future research should align AI literacy assessment with the educational goals and standards of kindergarten AI curricula to ensure that the assessment content aligns with the distinctive characteristics of teaching AI in kindergartens.

Secondly, we also found that the most commonly focused assessment construct was students' AI knowledge (Priya et al., 2022; Shamir & Levin, 2022). Most reviewed research assesses AI knowledge, while the remaining measures include AI ethics, affective aspects of AI, and the use of AI. Future research could explore a broader range of assessment constructs beyond AI knowledge, encompassing cognitive aspects and addressing affective components, AI utilization skills, and ethical considerations. This more comprehensive approach would provide a holistic understanding of students' engagement with AI.

Thirdly, employing diverse assessment tools for AI evaluations enables triangulation, facilitating a comprehensive assessment of participants' AI learning. Two main methods have been employed in research on assessing AI literacy, namely questionnaires

and surveys. Future studies may consider adding interviews and classroom observations as additional methods to evaluate participants' AI literacy.

Lastly, with the proliferation of AI, more researchers are focusing on AI. While conducting interventions related to AI literacy, many studies have reported demographic information. However, there needs to be more reporting on AI literacy assessment features related to students' demographic backgrounds. Future research may delve into demographic issues, such as item biases in AI literacy assessments, cultural responsiveness, and varied functional aspects of AI literacy assessments.

Conclusion and limitations

This review aids in mapping existing AI literacy assessments, including assessment tools, forms of assessment, and reliability and validity evidence. The results revealed an unequal development and application of AI literacy assessments in educational settings. Two-thirds of the research assesses K-16 students' AI literacy. We found that most studies used questionnaire, surveys and tests to assess participants' AI literacy in AI literacy assessment research, and most studies used computer-based tests, followed by paper-based tests. Results also showed that most studies used AI literacy assessment to assess participants' AI knowledge, followed by AI ethics. Finally, only a few studies reported evidence of the reliability and effectiveness of their assessments.

While the current scoping review extensively explores state-of-the-art AI literacy assessments, it is important to acknowledge several limitations inherent in the study. Despite numerous publications proposing new items for evaluating AI literacy, this review only included journal articles, excluding conference papers. Furthermore, as most studies reviewed turned out to predominantly reflect participants from Hong Kong, Mainland China, and the United States, there is a need for more research in other countries. Therefore, there is a need to shift more attention and research towards other regions worldwide.

Appendix

Overview of selected studies

	Participants' countries	Educational level/Adults	Teaching methods	Assessment tools	Forms of assessment tools	Assessment dimensions	Reliability	Data analyse methods	Software
Vartiainen et al. (2020)	Finland	Kindergarten and primary	Learning by design	Video recordings and interviews	Video-based and oral-based	Children's interaction with a Teachable Machine	/	Qualitative content analysis	/
Chiu (2021)	Hong Kong	Secondary	Learning by design	Questionnaire	Paper-based	AI knowledge, attitude, and motivation toward learn AI	0.88–0.92	Paired *t*-tests and analyses of covariance (ANCOVA)	SPSS
Priya et al. (2022)	India	Secondary	Game-based learning	Survey	Paper-based	Three machine learning concepts: Supervised Learning, Gradient Descent and K-Nearest Neighbor (KNN) Classi cation	/	Descriptive Statistics	/
Shamir and Levin (2022)	Israel	Primary	Learning by design	Questionnaire	Computer-based	Machine Learning (ML) process and interest in learning ML	/	/	/
Dai et al. (2023b)	Mainland China	Primary	Learning by design	Pre- and post-interviews	Oral-based	AI knowledge	/	Nonparametric statistics—Wilcoxon signed-rank test and the Wilcoxon rank-sum test	SPSS
Wood et al. (2021)	USA	University	/	Survey	Computer-based	Attitudes toward AI	/	Descriptive statistics (means, standard deviations, frequencies, and percentages) and Chi-square	SAS

	Participants' countries	Educational level/Adults	Teaching methods	Assessment tools	Forms of assessment tools	Assessment dimensions	Reliability	Data analyse methods	Software
Laupichler et al. (2023a)	Various countries	Adult	/	Questionnaire	Computer-based	AI knowledge	/	The mean, median, and standard deviation	SPSS
Choi and Park (2021)	Korea	Primary	Gamification	Survey	Paper-based	AI knowledge	/	/	/
Zhao et al. (2022)	Mainland China	Teachers	/	Survey	Computer-based	AI knowledge, applying AI, evaluating AI application, and AI ethics	0.963 and 0.967	Exploratory factor analysis and KMO	AMOS
Laupichler et al. (2023b)	Various countries	Adult	/	Survey	Computer-based	AI knowledge	0.93	EFA	R
Laupichler et al. (2022a, b)	Germany	University	Flipped classroom	Questionnaire	Computer-based	AI knowledge	0.88	A repeated measures ANOVA	SPSS,R, Python
Dai et al. (2023a)	Mainland China	Primary	Learning by design	Drawing task, assessment task, and Scratch project	Paper-based and computer-based	AI knowledge, AI skills, and AI ethics	0.76	Independent Sample *t*-test	SPSS
Laupichler et al. (2023c)	Germany	University	Learning by design	Questionnaire	Paper-based	Attitudes toward AI and AI knowledge	/	*t*-test	SPSS
Kong et al. (2023a)	Hong Kong	University	Learning by design	Tests, surveys, reflective writings and interviews	Paper-based and Voice-based	AI concepts and AI ethics	0.66–0.93	Paired *t*-test	SPSS
Kong et al. (2023b)	Hong Kong	Secondary	Learning by design	Pre- and post-course tests, surveys and self-reflective writing	Paper-based	AI-related concepts and AI ethics	0.74–0.97	Wilcoxon signed-rank test	SPSS

	Participants' countries	Educational level/Adults	Teaching methods	Assessment tools	Forms of assessment tools	Assessment dimensions	Reliability	Data analyse methods	Software
Kong et al. (2022)	Hong Kong	University	Learning by design	Conceptual tests, surveys and self-reflective writing tasks	Paper-based	AI concepts, machine learning and deep learning	0.66–0.93	Paired *t*-test	SPSS
Kong et al. (2021)	Hong Kong	University	Learning by design	AI concepts test and AI literacy survey	Computer-based	AI concepts	0.66–0.93	Paired *t*-test	SPSS
Mertala et al. (2022)	Finland	Primary	Learning by design	Survey	Computer-based	Students' conceptions of AI	/	/	/
Estevez et al. (2019)	Spain	Secondary	Learning by design	Questionnaire	Paper-based	Basic AI knowledge	/	/	/
Zhang et al. (2024)	USA	Secondary	Learning by design	Survey, presentation and interviews	Computer-based and oral-based	AI concepts, interest in AI, and the ethical and social implications of AI	0.94	Paired *t*-test results	SPSS
Wang et al. (2023)	Mainland China	Adult	/	Questionnaire	Computer-based	Awareness, use, evaluation, and ethics	0.73–0.78	CR and AVE	SPSS
Lin et al. (2023)	Mainland China	Secondary	/	Survey	Paper-based	AI knowledge and skills	0.9	Structural equation modelling	AMOS
Henry and Collard (2021)	Belgium	Primary and Secondary	Learning by design	Test, observations and recording	Paper-based	AI knowledge	/	/	/
Cardon et al. (2023)	USA	Adult	/	Survey	Computer-based	AI application, authenticity, accountability, and agency	0.79–0.96	/	/
Ng et al. (2022)	Hong Kong	Primary	Digital story writing	Interviews and writing artefacts	Oral-based and computer-based	AI concepts	/	/	/

	Participants' countries	Educational level/Adults	Teaching methods	Assessment tools	Forms of assessment tools	Assessment dimensions	Reliability	Data analyse methods	Software
Ng et al. (2023)	Hong Kong	Secondary	Learning by design	Test, questionnaire, presentation and interviews	Computer-based and oral-based	AI knowledge, use AI, and AI ethics	0.88	*t*-test	SPSS
Hornberger et al. (2023)	Germany	University	/	Test	Computer-based	AI knowledge	0.82	IRT	Rasch
Su and Yang (2023a)	Hong Kong	Kindergarten	Learning by design	Test	Oral-based	Basic AI concepts	0.9	*t*-test	SPSS
Su and Yang (2023b)	Hong Kong	Kindergarten	Learning by design	Questionnaire	Oral-based	Perceptions and attitudes toward AI	0.9	ANOVA	SPSS
Su (2024)	Hong Kong	Kindergarten	Learning by design	Test	Oral-based	Basic AI concepts and knowledge	0.78	IRT	R
Carolus et al. (2023)	Germany	Adults	/	Questionnaire	Computer-based	Apply AI,AI knowledge, and AI Ethics	0.66–0.93	CFA	R
Ng et al. (2023)	Hong Kong	Secondary	Learning by design	Questionnaire	Computer-based	Affective, behavioural, cognitive and ethical dimensions	0.93	CFA	R
Ding et al. (2024)	United States	In-service and pre-service teachers	/	Test	Computer-based	AI knowledge, AI ethics	/	CFA and IRT	R
Laupichler et al. (2024)	Germany	Adult	/	Questionnaire	Computer-based	AI knowledge and attitudes towards AI	0.53–0.94	CFA	R
Yuan et al. (2024)	Taiwan	13–70 years old	/	Survey	Computer-based	AI knowledge and AI ethics	0.75–0.87	CFA	Mplus
Zhang et al. (2024)	United States	Secondary	/	Test	Computer-based	AI concepts	0.73	IRT	R

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Jiahong Su obtained her PhD from the Faculty of Education at the University of Hong Kong, Hong Kong. Her research interests focus on AI in education, AI technologies, and AI literacy. She has published over 30 papers in international journals. Her research has been published in top international journals, including *Computers & Education, British Journal of Educational Technology, Educational Technology Research and Development* and other prestigious journals. She is an editorial board member for the *Cambridge Elements series* on Generative AI in Education and *Discover Education journals*, as well as an early career researcher editorial board member for the *British Journal of Educational Technology*.

Xinyu Chen is a Ph.D. candidate in the Faculty of Education in the University of Hong Kong. She holds a Master Degree (M.Ed. with distinction) in HKU, as well as a Bachelor's degree (B.Ed.) in EdUHK. She has various learning and teaching experiences in Hong Kong, mainland China and Europe, which brings her multicultural insights into different learning and living styles. Her research interest covers broad topics of Gamification in Education, Game-based Learning, Technology Education and Artificial Intelligence (AI) Education.

Samuel Kai Wah Chu is a Professor in the Department of Health Sciences, School of Nursing and Health Studies at the Hong Kong Metropolitan University. He has obtained 2 PhDs in Education – one focusing on e-Learning from University College London, Institute of Education (2017) and another one focusing on Information and Library Science from HKU (2005). His areas of expertise include AI literacy, gamified learning, 21st Century Skills, and Social Media in Education. He has been involved in over 70 research projects with a total funding of US$ 9,391,342. He has published more than 400 articles and books, with over 100 of them appearing in international academic journals. Furthermore, Dr. Chu is the Co-Founder and Co-Editor for the journal Information and Learning Sciences. He is also a Member of the Humanities and Social Sciences Panel of the Research Grants Council of HK. He has received many awards including the Faculty Outstanding Researcher Award in 2013, Faculty's Knowledge Exchange Award in 2016 and Excellent Health Promotion Project Award from Food and Health Bureau in 2017. He's ranked among the top 2% of scholars in 2 research areas: Information & Library Sciences and Education (PLoS Biol 18(10), 2020, a study by Stanford University), and achieved an h-index of 39 over the years. He has been a Consultant for UNESCO Bangkok, Education Bureau, Oxford University Press, Pearson Education Asia, and is currently serving as the ASIS&T Director of Chapter Assembly (2020–2021). He is also an Advisor for EdTech start-ups at HK Science and Technology Parks and the Founding Chairman of Academy 22 Education for All Foundation. One of his advisory companies is now valued at $0.3 billion HK.

Xiao Hu is an Associate Professor in the Faculty of Education at the University of Hong Kong (HKU). Her research in technology-enhanced learning and learning analytics has been funded by the Research Grant Council, Environment Conservation Fund, and Quality Education Fund of Hong Kong S.A.R. She is the coordinating co-chair of the Big Data in Education and Learning Analytics (BDELA) track in the IEEE International Conference on Advanced Learning Technologies (ICALT) and an associate editor of Information Processing & Management (IP&M).